

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech (EE) Examination

April-May 2018

EE247 Transformers and Induction Machines

Date: 07.05.2018, Monday

Time: 10.0 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the right answer for the following statements.

[07]

a. Which of the following is minimized by laminating the core of a transformer

- | | |
|--------------------|----------------------|
| a. Hysteresis loss | b. Eddy current loss |
| c. Heat loss | d. All of the Above |

b. The function of breather in a transformer is

- | | |
|---|-----------------------------------|
| a. To provide oxygen to the cooling oil | b. To provide cooling air |
| c. To arrest flow of moisture when outside air enters the transformer | d. To filter the transformer oil. |

c. Iron loss in a transformer occurs in

- | | |
|-------------------|--------------|
| a. Core | b. Winding |
| c. Insulating oil | d. Main body |

d. Open circuit test of a transformer gives

- | | |
|---|----------------------|
| a. Hysteresis loss | b. Eddy current loss |
| c. Sum of hysteresis loss and eddy current loss | d. Copper loss |

e. The efficiency of a transformer does not depends on

- | | |
|-----------------|---------------------|
| a. Current | b. Load |
| c. Power factor | d. All of the above |

f A transformer can have zero voltage regulation at

- a. Leading power factor
- b. Zero power factor
- c. Lagging power factor
- d. Unity power factor

g Three-to-three phase transformer connection possible for parallel operation is

- a. Δ -Y to Δ -Y
- b. Δ - Δ to Δ -Y
- c. Y-Y to Δ -Y
- d. Δ -Y to Y- Δ

Q – 2.a A 10 KVA, 500/250V, 50Hz single-phase transformer has a net area of cross section 90 cm² and maximum flux density is 1.2T. Calculate the number of turns on both the primary and secondary. **[04]**

Q – 2.b Draw equivalent circuit and phasor diagram of a transformer on load and therefrom do the power flow analysis. **[05]**

OR

Q - 2.b A 6,600/400 V, single-phase 600KVA transformer has 1,200 primary turns. Find (1) transformation ratio (2) secondary turns (3) voltage per turn (4) secondary current when it supplies a load of 400kw at 0.8 power factor lagging. **[05]**

Q – 2.c What is polarity test? Why and how this should be performed in a single phase two winding transformer? Explain clearly with the help of connection diagram. **[05]**

OR

Q – 2.c What are the different losses in the transformer? How do they depend on voltage and frequency? **[05]**

Q - 3 a. Explain the open-delta connection of a 3-phase transformer. What are the applications of this system? **[07]**

Q - 3 b. Describe sumpner's (back to back) test used for transformers. Indicate its advantages **[07]**

OR

Q - 3 a. In what way is all day efficiency different from conventional efficiency? How is all day efficiency calculated? **[07]**

Q - 3 b. A 20KVA, 250/2,500V, 50Hz, 1-phase transformer gave the following test results: **[07]**

OC (LV Side)	250V	1.4A	105W
SC(HV Side)	120V	8A	320W

Find the parameters and draw the equivalent circuit of 1-phase transformer as referred to low voltage side.

SECTION – II

- Q – 4** Draw and explain power flow diagram of induction motor. [07]
- Q – 5.a** Derive torque equation of induction motor for starting and running condition. [05]

OR

- Q – 5.a** A 3-phase induction motor is wound for 6 poles and is supplied from 50Hz, AC system. [05]
Calculate (1) The synchronous speed (2) The speed of rotor when the slip is 5% (3) The frequency of rotor currents when rotor runs at 800 r.p.m and (4) The frequency of rotor current at standstill.
- Q – 5.b** Describe Cogging and Crawling phenomenon in Induction motor? [04]
- Q – 5.c** Draw and explain the working of D.O.L starter. [05]

OR

- Q – 5.c** A 6-pole, 50Hz, 3-phase induction motor has an effective rotor resistance of 0.08 ohm per phase. If its stalling speed is 750 r.p.m, find the rotor reactance per phase. Also find the external rotor resistance to be included to obtain maximum torque at starting. [05]
- Q – 6.a** State various methods for speed control of induction motor from stator side and explain any one of them. [07]
- Q – 6.b** Explain the double field revolving theory of operation of single phase induction motor. [07]

OR

- Q – 6.a** Explain capacitor start motor in context with its construction, operation, advantages and applications. [07]
- Q – 6.b** Draw the circle diagram for a 3-phase, 6 pole, 50Hz, 415V star connected induction motor from the following data (Line Values): [07]

No-load Test	$V_0 = 415V$	$I_0 = 9A$	$W_0 = 1200W$
Full-load test	$V_{sc} = 200V$	$I_{sc} = 50A$	$W = 6900W$

The stator loss at standstill is 60% of total copper losses and full load current is 32A.

From the circle Diagram, determine the following performance parameters:

1. Full load power factor
2. Power output
3. Full load slip
4. Total losses
5. Motor efficiency
6. Full load torque
7. Maximum torque
